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MILITARY SPECIFICATION

DISTRIBUTORS, WATER, TANK-TYPE: TRUCK MOUNTED

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a self-loading water distributor with equipment for dispensing water through spray attachments.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

ZZ-H-451	- Hose, Fire, Woven-Jacketed Rubber or Latex or Rubber-Coated-Fabric-Lined, with Couplings.
ZZ-H-561	- Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water Suction and Discharge.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-T-740	- Trucks; 5 Ton, 6X6, Military Design, M39, M39A1, M39A2 and M809 Series.
MIL-D-771	- Distributors, Bitumen and Water, Packaging of.
MIL-C-1283	- Can, Gasoline, Military, 5-Gallon.
MIL-S-1285	- Spout, Can, Flexible.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2638	- Lacquer, Vinyl Resin, Gasoline and Water Resistant.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-L-4387	- Lubrication Dispensing Equipment Accessories.
MIL-A-8421	- Air Transport Requirement, General Specification.
MIL-S-8660	- Silicone Compound, NATO Code Number G-354, Metric.
MIL-P-10360	- Pumps, Centrifugal; 4 and 6 Inch, Gasoline-Engine-Driven and Diesel-Engine-Driven, for Water.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-N-12314	- Nozzles, Fire Hose, Water: Spray and Stream.
MIL-C-13486	- Cable, Special Purpose, Electrical: Low-Tension, Heavy Duty, Single-Conductor and Multiconductor.
MIL-I-15126	- Insulation Tape, Electrical, Pressure Sensitive Adhesive and Pressure Sensitive Thermosetting Adhesive.
MIL-V-16721	- Valves, Foot, Low Pressure, Cast-Iron (for Water).
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354, Metric.
MIL-C-24066	- Clip, Component, Non-electrical; General Specification For.

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MIL-C-24066/2 - Clip, Solid, Spring Tension.
MIL-B-43436 - Band Markers, Blank, General Specification For.
MIL-B-43436/1 - Band Markers, Blank: Single Crimp Style Band.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-52404 - Connections: Hose, Fire and Water.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-209 - Slinging and Tiedown Provision for Lifting and Tying Down Military Equipment.
MIL-STD-810 - Environmental Test Methods and Engineering Guideline.
MIL-STD-838 - Lubrication of Military Equipment.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1474 - Noise Limits for Army Materiel.
MS21322 - Clamp, Loop-Aluminum, Plain and Cushioned.
MS21333 - Clamp, Loop-Steel, Plain and Cushioned.
MS21334 - Clamp, Loop-Double, Cable or Tube, Steel, Plain and Cushioned.
MS27142 - Connector, Plug, Electrical - Pin Contact No. 12, 14 and 16 A.W.G., Waterproof.
MS27143 - Connector, Plug, Electrical: Socket Contact, No. 12 A.W.G., Waterproof.
MS27144 - Connector, Plug, Electrical: Socket Contact, No. 14 and 16 A.W.G., Waterproof.
MS27145 - Connector, Plug, Electrical: No. 14 A.W.G., Double Contact, Waterproof.
MS27146 - Connector, Receptacle, Electrical: Double Contact, Waterproof.
MS27147 - Adapter, Connector - 3-Way No. 12, 14, and 16 A.W.G., Waterproof.
MS27148 - Contact; Pin, Electrical Connector: No. 12, 14, and 16 A.W.G.
MS33800 - Connector, Receptacle, Electrical-Pin Contact, No. 12, 14 and 16 A.W.G., Waterproof.
MS35387 - Reflector, Indicating, Clearance.
MS35422 - Lampholder, Light, Marker, Clearance.
MS35423 - Light, Marker, Clearance - Service.
MS35424 - Light, Marker, Clearance - Blackout, 24 Volt.

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MS39153 - Coupling, Quick Disconnect, 3- and 4-Inch for Construction Equipment.
MS51330 - Stop Light - Tail Light, Vehicular - 24 Volt, Blackout Tail, Blackout Stop.

HANDBOOKS

MILITARY

MIL-HDBK-113 - Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DOT)

DOT Motor Carrier Safety Regulations.

(Application for copies should be addresssed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402).

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1 - Structural Welding Code - Steel.

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(Application for copies should be addressed to the American Welding Society Inc., 550 N.W. LeJune Road, Miami, FL 33135.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook	
SAE J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
SAE J561	- Electrical Terminals - Eyelet and Spade Type.
SAE J562	- Nonmetallic Loom.
SAE J759	- Lighting Identification Code.
SAE J910	- Hazard Warning Signal Switch.
SAE J945	- Vehicular Hazard Warning Signal Flasher.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

STEEL STRUCTURE PAINTING COUNCIL (SSPC)

SSPC-SP6, Commercial Blast Cleaning

(Application for copies should be addressed to the Steel Structure Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The distributor shall consist of a tank, an engine-driven water pump, a spraybar and necessary controls, piping and safety and warning devices. The distributor shall be mounted on a military truck chassis as specified herein. The overall width of the distributor shall be not more than 96 inches (244 cm).

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.6). Any change or deviation of the distributor from the approved first article during production will

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be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish a distributor conforming to this specification.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. All sheet steel gages specified herein shall be United States Standard Gage.

3.3.1 Material deterioration prevention and control. The distributor shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the distributor may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contactor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the distributor may be newly fabricated from recovered materials to the maximum extent practicable, provided the distributor produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the distributor.

3.4 Tank. Each tank shall have a capacity of not less than 1,000 gallons (3,785 liters), and shall be an elliptical steel cylinder. The shell, baffles, heads, manhole, vents and outlets shall be fabricated from steel or type 304 stainless steel. If steel is used, the interior of the tank surface shall be free of weld spatter, oil and grease prior to sandblasting. The interior metal surfaces shall be sandblasted in accordance with SSPC-SP-6 with a surface profile not to exceed 3 mils. The interior surfaces shall be treated with a self priming non-toxic coal tar material suitable for water tanks, within 8 hours of sandblasting. Coating thickness shall be not less than 20 mils dry film thickness average, with a minimum of 16 mils. All areas less than 16 mils shall be recoated to achieve 20 mils. If top coating cannot be applied within the 8 hour period, shop prime metal surfaces to a dry film thickness of 2 mils. Steel sheet used in the tank cylinder shall be 10 gage. Steel sheet used in the heads shall be 8 gage. The heads shall be flat and reinforced, concave or convex, and all seams shall be welded and watertight. Each tank shall have

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transverse steel surge plates made of 10 gage sheet steel, welded in place. The spacing between surge plates and between surge plates and tank ends shall not be more than 48 inches (1.22 m). The surge plates shall be the full size of the tank cross section, and shall be dished or reinforced to withstand the surge pressure. Holes in the surge plates shall be staggered. The surge plates shall permit free flow of water at the bottom, venting at the top, and access to all sections of the tank. Each tank shall be provided with a manhole not less than 21 inches (53.34 cm) in diameter. The manhole shall be located at the top of the tank and shall be equipped with a hinged, watertight, quick-clamp-type cover. A 3-inch (7.62 cm) unrestricted opening for overflow and for venting to the atmosphere shall be provided. The 3-inch (7.62 cm) opening shall be protected against surge discharge, and shall be located not more than 2 feet (61 cm) from the transverse centerline of the tank. Each tank shall be provided with a sump to allow complete draining through the pump. The sump shall be provided with a separate drain so that all the water can be drained from the sump when desired.

3.4.1 Tank gage. A tank gage shall be provided. The gage shall be calibrated in gallons from zero to tank capacity in increments of not more than 25 gallons (88.63 liters). The gage shall have an accuracy such that the indicated amount of water shall be within 25 gallons (88.63 liters) of the actual amount of water in the tank. The gage shall be protected from breakage and shall be readable from the operator's platform.

3.5 Pump unit. Each distributor shall be provided with a pump and engine unit conforming to MIL-P-10360, type I, size 4, class A, except that a three-way fuel valve shall not be required. A tank mounted fuel level indicator shall be provided on the pump engine fuel tank. The indicator shall be readable by the operator when in his normal operating position. The pump unit shall be mounted at the rear of the distributor and shall be accessible for service and maintenance. The pump engine cranking motor shall be powered from the truck batteries. The pump engine exhaust shall be routed to discharge beneath the truck chassis. The exhaust system shall be adequately supported or clamped to prevent damage by vibration. Where the exhaust is located so as to be exposed to operating personnel, it shall be fully guarded or insulated. The routing of the exhaust system shall permit the removal of the pump-engine combination for maintenance purposes without any cutting.

3.5.1 Pump fittings and connections. The necessary pipe, fittings and controls shall interconnect the pump to the distributor in a manner such that the following operations will be possible:

a. Pump water from an outside source, such as lakes, ponds, or streams, into the distributor tank.

b. Pump water from the distributor tank to the spraybar.

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c. Pump water from the distributor tank to a 1.5-inch (3.81 cm) hose outlet without water entering the spraybar.

d. Pump water from an outside source, without water entering the spraybar or tank, to a 1.5-inch (3.81 cm) hose outlet.

A removable, galvanized, full-opening-type strainer with perforations smaller than the spray nozzle opening shall be placed in the line ahead of the pump so as to filter all water from the tank or outside source prior to pump entry. The strainer shall be removable without removing any fittings or connections other than the 4-inch (10.16 cm) inlet dust cover. Piping from the pump to the spraybar shall include a 1.5-inch (3.81 cm) hose outlet with external threads (NHT). The suction side of the pump shall be furnished with a 4-inch (10.16 cm) coupling conforming to MS39153 (faucet end) for connecting the suction hose. The 4-inch (10.16 cm) inlet and the 1.5-inch (3.81 cm) outlet shall be provided with a dust cover secured by a sash chain.

3.6 Pressure gage. A pressure gage shall be connected to the discharge side of the pump to indicate pressure in the line. The gage shall be calibrated in 5-pound (2.27 kg) increments from 0 to not less than 110 percent of the maximum working pressure. The gage shall be readable from the normal operator's position.

3.7 Spraybar. The distributor shall be furnished with a spraybar equipped with spray nozzles. The spraybar shall consist of an 8-foot (2.44 m) center section, with provisions to add extensions to each end of the spraybar to make a total length of not less than 16 feet (4.88 m). The extension sections shall be in 1-foot (30.48 cm) and 2-foot (60.96 cm) lengths. Each 16-foot (4.88 m) spraybar shall be provided with four 1-foot (30.48 cm) extension sections. The two ends of the center section of the spraybar and the extension sections shall be provided with a snap-over-center type coupling device to connect extension sections to center section and to connect each extension section together. The coupling device shall permit connection and disconnection by hand without the use of tools. The coupling device shall permit connection of extension sections in only one position to positively maintain nozzle alignment. The caps on each end of the spraybar shall be connected by use of the coupling device. The extension-section connections and the end-cap connections shall not leak during any operation specified herein. When the extension sections are attached to the spraybar, provisions shall be made to reduce the length of the spraybar to 8 feet (2.44 m) by folding the extended section. The spraybar shall lock in both the folded and extended positions. The flow of water to the folded section shall be automatically cut off when the bar is in the folded position. Supports shall be provided for all bar lengths to prevent damage or deformation to the spraybar during operation or highway travel. The ends of the bar shall swing horizontally without damage to the bar when the extended sections strike an obstruction. The bar shall be equipped with brass slotted nozzles spaced not greater than 6 inches (15.24 cm) on centers. The threaded portion of the nozzles shall not be less than 0.675 inch (1.71 cm) in diameter

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and shall be attached to threaded bosses that are welded on, or built into, the spraybar. The nozzles shall not extend beyond the inside diameter of the spraybar. The nozzles shall be mounted in such a way that allows all nozzles to be adjusted by a control which shall be operable from the operator's platform. The bar shall be vertically adjustable by means of mechanical linkages and shall be operable from the operator's platform. Vertical adjustment shall be not less than 10 inches (25.5 cm). The spraybar shall be supported to prevent swaying at any vertical adjustment during operation or travel. A means shall be available to drain all water from the spraybar.

3.7.1 Spraybar clearance. The spraybar shall be located so that the angle formed between the ground plane and a plane passing through the lowest point of the bar assembly (bar in raised position) and the ground contact points of both rear tires shall not be less than 20 degrees when the tank is full of water.

3.7.2 Application and distribution. The width of distribution shall be 4 feet (1.22 m), 8 feet (2.44 m), and from 8 feet (2.44 m) through the full width of spraybar in increments of 1 foot (30.48 cm). The rates of application shall be from 0.1 gallon through 2.0 gallons per square yard (0.475 liters through 9.50 liters per square meter) in increments of 0.1 gallon per square yard (0.475 liters per square meter). The truck-mounted distributor shall apply water through the spraybar to the surface being sprayed at all combinations of specified widths and rates. The accuracy of distribution over a given area shall be within plus or minus 5 percent. The application shall be accomplished without streaking or puddling, and spray from each nozzle shall overlap spray from each adjacent nozzle by not less than 50 percent at any spraybar height within the 10-inch (25.4 cm) vertical, adjustable range. The forward speed for any distribution rate shall be not less than 88 feet (27 m) per minute (fpm) and shall be not more than 1,350 fpm (411.48 mpm).

3.8 Controls. All operating controls shall be located within reach of 5th percentile female through 95th percentile male operators while standing on the operator's platform. All water control valves shall be of the lubricated type or shall have internal parts coated with a permanent type coating to prevent corrosion and binding of the valve. Controls shall be clearly identified for their intended operation, and function on permanently affixed plates of corrosion resistant material. All rotary valve controls shall be provided with double-ended arrows showing directions of operation and labeled at each end to indicate the functional result (i.e. open, close, etc.). Controls shall permit the following operations:

a. Fill the distributor tank from an outside souce using the pump.

b. Pump water from the distributor tank to either or both halves of the spraybar.

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c. Pump water from the distributor tank to a 1.5-inch (3.81 cm)-diameter connection for hose attachment without water entering the spraybar.

d. Empty the distributor tank, by gravity through the spraybar.

e. Pump water from an outside source, without water entering the spraybar or tank, to a 1.5-inch (3.81 cm) hose outlet.

An operating instruction plate shall be provided describing step-by-step procedures required to perform all operations specified herein, and shall include a flow and valve diagram. The plate shall be permanently affixed to the distributor so as to be readable from the operator's platform.

3.9 Operator's platform. A platform for the operator shall be provided at the rear of the distributor. The platform shall extend the full width of the distributor and shall be provided with nonskid metal decking. The platform shall be equipped with a rack for holding the discharge hose. The platform shall provide access to all distributor controls located at the rear of the distributor. Steps shall be provided for access to the platform from the ground, and shall be fabricated of nonskid metal decking. The platform shall be fabricated of open expanded metal grating.

3.10 Speed-and-distance-indicating device. A speed-and-distance-indicating device shall be furnished and shall provide the following:

a. Traveling-speed indicator reading in feet per minute from 0 to 1,500, graduated in increments of not more than 20. The accuracy of the indicator shall be such that the indicated speed shall be within 5 percent of the actual speed.

b. Cumulative-distance indicator reading to 999,999 feet (304799.69 m).

c. Trip indicator reading to 9,999 feet (3047.70 m), resettable to zero.

The accuracy of the indicator shall be such that the indicated distance traveled shall be within 5 percent of the actual distance traveled. The indicators shall be operated by a rubber-tired, ground-contact wheel. The wheel shall be lowered to the ground contact position and raised to the travel position by a control in the cab of the truck. The wheel shall maintain contact with the ground at all times during spraying operations. The clearance when the wheel is locked in the raised position shall be not less than the inter-axle ground clearance of the truck. The indicators shall be located in the cab of the truck visible to the truck operator when he is in his normal operating position.

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3.11 Lubrication.

3.11.1 Lubricants. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a qualified products list (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.11.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III and shall be located in a protected position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a straight hydraulic coupler and a flexible extension conforming to MIL-L-4387, type III, class 2.

3.11.3 Pressure-release device. An automatic pressure-release device shall be provided where pressure lubricating equipment will damage grease seals or other parts.

3.11.4 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with grease conforming to MIL-G-10924 or MIL-G-23827, as applicable. The distributor shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not always compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.11.5 Oil lubrication. Unless otherwise preserved for packaging, all components shall be filled to the operating level with grades of military oils conforming to MIL-L-2104, designated for use in the temperature range at the shipping destination. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricants used.

3.12 Transportability. When unmounted, or mounted on a military truck chassis, the distributor when empty, partially filled, and full, shall be transportable by highway. When empty the distributor shall be transportable by rail, water and air. Except for width, the truck mounted distributor shall meet highway limits in CONUS. Dimensions of this configuration on a 50-inch (127 cm) high railroad car shall be within the outline diagram for single loads, without end overhang, on open-top cars as shown in figure 1. Both mounted and unmounted configurations shall be capable of completing a rail impact test in accordance with MIL-STD-810, without having damage or permanent deformation. Both configurations shall meet the requirements of MIL-A-8421 for air transport in Military Airlift Command aircraft.

3.12.1 Lifting attachments. Permanently affixed lifting attachments that enable the unmounted distributor to be lifted to its normal travel position shall be provided. Each attachment shall meet the requirements specified in

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MIL-STD-209, for type II equipment, class 1 or 3 provisions. Lifting attachments may also be used as tiedown attachments when such attachments meet the requirements specified in 3.12.2.

3.12.2 Tiedown attachments. Permanently affixed attachments that permit fastening the unmounted distributor to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall meet the requirements specified in MIL-STD-209, for type II equipment, class 2 or 3 provisions.

3.13 Accessories. A suction-hose assembly and a discharge-hose assembly shall be provided with each distributor with provisions for carrying the accessories on the distributor.

3.13.1 Suction hose assembly. The suction hose shall consist of three 10-foot (3.048 m) sections of 4-inch (10.16 cm) hose conforming to ZZ-H-561, grade A, type I. Two sections of the hose shall be fitted with 4-inch (10.16 cm) couplings conforming to MS39153 (hose-end on one end, faucet-end on the other end). Each faucet-end shall be provided with a dust cover secured by chain. One double adapter shall be furnished. The third section of the hose shall be fitted on one end with a hose-end, quick-disconnect coupling, and on the other end with a 4-inch (10.16 cm) pipe nipple with NPT threads. A 4-inch (10.16 cm) foot valve conforming to MIL-V-16721 shall be provided for use with the suction hose.

3.13.2 Discharge hose assembly. The discharge hose shall be 1.5 inches (3.81 cm) inside diameter, 25 feet (7.62 m) long, and shall conform to ZZ-H-451, type III, class B. A nozzle conforming to MIL-N-12314, type III, class A, 1.5-inch (3.81 cm) with 5-inch (12.7 cm) tip opening, shall be provided. A 2.5-inch (6.35 cm) by 1.5-inch (3.81 cm) reducing connection conforming to MIL-C-52404, type XV, class A, shall be provided to connect the nozzle end of the hose to a fire hydrant. Fixtures with hose threads shall be mounted in the toolbox to provide safe storage for the connector and nozzle.

3.14 Threads. Threads shall conform to FED-STD-H28.

3.15 Repair parts and maintenance tools. Such repair parts, operating tools and maintenance tools as are specified shall be furnished (see 6.2).

3.16 Toolbox. The distributor shall be provided with a lockable toolbox made of 14-gage sheet steel. The toolbox shall have an overlapping hinged lid which prevents the entry of rain and dirt. The lid shall be held closed with two hook-type fasteners. The toolbox shall be of a size to hold the tools specified in 3.15, the fixtures, with connector and nozzle, specified in 3.13.2, and the spraybar extension sections specified in 3.7. Hold-downs shall be provided for the spraybar sections. The toolbox shall have drain holes which prevent the collection of water.

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3.17 Mounting. The distributor shall be mounted on a Government-furnished military 5-ton, 6 by 6, truck chassis, specific model shall be specified by the contract officer (see 6.2.1), with personnel heater and defroster, slave receptacle kit and without winch in accordance with MIL-T-740 (see 6.3). The appropriate chassis drawings will be furnished by the Government (see 3.28). When specified (see 6.5), all distributors, except the first article model, shall be furnished unmounted. The distributor shall be mounted on the truck chassis with not less than four mounting devices per side. Each mounting device shall consist of one U-bolt, or two twin studs, with tie plates and nuts and washers as required. Hardwood filler blocks shall be placed in chassis sill channels at each mounting point. Each filler block shall be retained in the channel by use of keeper straps or by use of grooves around mounting bolts or studs. Each filler block shall be longitudinally centered on the mounting bolt or stud. A hardwood breaker strip shall be installed between the longitudinal distributor sills and longitunial chassis sills, and shall run the full length of the chassis sills. Additional mounting devices of a type required to restrict transverse and longitudinal movement between distributor and chassis shall be installed. The truck chassis frame shall not be welded to or extended. The distributor shall be mounted without drilling holes in the chassis, unless prior approval is obtained from the contracting officer. The alinement between the truck chassis and distributor shall be maintained, and the mounting devices shall not be permanently deformed or broken when the fully loaded, truck-mounted distributor is subjected to travel over paved and gravel roads and subjected to sudden stops when traveling at forward speeds up to 10 mph (6.21 kmph) and at maximum, high-range reverse speeds. When unmounted distributors are furnished, all components required for mounting and an illustrated step-by-step mounting procedure shall be supplied with each distributor.

3.18 Treatment and painting. The portions of the distributor normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage green 383 conforming to MIL-C-46168. When specified (see 6.2.1), the distributor shall be overcoated in accordance with the Government furnished camouflage patterns, with a MIL-C-46168 top coat of the colors specified in the camouflage patterns, except the first article may be camouflage green 383.

3.18.1 Camouflage pattern data. When specified (see 6.2.1), the contractor shall provide camouflage pattern data.

3.19 Electromagnetic interference. The electromagnetic interference emission characteristics of the distributor shall conform to the requirements of SAE J551.

3.20 Distribution chart. Two distribution charts shall be provided with each distributor. The charts shall provide the following information: (a) the pressure (pump gage); and (b) the travel speed in feet-per-minute required

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to distribute water at all rates specified herein for spraybar-lengths of 4 feet (1.22 m), 8 feet (2.44 m), and from 8 feet (2.44 mm) through the full width of the spraybar in increments of 1 foot (30.48 cm). The charts shall be protective coated with clear plastic material.

3.21 Identification marking. Each distributor shall be identified in accordance with MIL-STD-130. Marking shall be applied to the distributor on identification plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous, protected location.

3.21.1 Shipping data plates. Each distributor shall be provided with two shipping data plates conforming to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. One plate shall indicate shipping data for the unmounted distributor and shall be entitled "Shipping Data for Distributor". The other plate, showing distributor shipping data, shall contain a warning to indicate that distributor lifting attachments are not to be used to lift the truck-mounted distributor. One plate shall show the silhouette of the distributor in transport position, indicating the center of gravity and the location and capacity of the lifting and tiedown attachments. The other plate shall show the silhouette of the truck-mounted distributor in transport position, indicating the center of gravity. The recommended tiedown arrangement shall also be shown on the plates. The plates shall be attached by screws, bolts, or rivets in a conspicuous, protected location.

3.21.2 Instruction plates. Each distributor shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the distributor. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by bolts, screws, or rivets in a conspicuous, protected location, and readable from the operator's normal position.

3.22 Stenciling. The gross vehicle weight of the truck-mounted distributor shall be applied on each side of the distributor. When the distributors are furnished unmounted, the gross vehicle weight of the truck-mounted distributor shall not be applied to the distributor, but shall be furnished with mounting procedures (see 3.17). The prescribed tire pressure shall be stenciled near each tire. Lifting and tiedown attachments shall be marked "LIFT DISTRIBUTOR HERE" and "TIEDOWN DISTRIBUTOR HERE" as applicable.

3.23 Electrical system. The distributor shall be equipped with a waterproof, 24-volt, vehicular electrical system for service and blackout night travel. All components and wiring shall be so located, and guarded, as to prevent damage, and permanent deformation, and to provide maximum protection from shock and vibration. The components and wiring shall also be accessible

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for replacement without disturbing permanently installed housings and guards. The electrical system shall be in accordance with table I. Item quantities, that are not specified shall be furnished as required (AR) for each distributor.

TABLE I. Electrical system.

Description	Requirement	Quantity
Service lamps: Clearance, side-marker, and identification	MS35423	14 ea., min.
Blackout lamps: Clearance and sidemarker	MS35424	8 ea.
Installation	Horizontal, using information on MS35422	AR
Service turn signal lamps, front	Commercial, SAE I (SAE J759), pedestal mount, waterproof	2 ea.
Service combination lamps, rear	Commercial, SAE STI (SAE J759), recess or flush mount, waterproof	
Blackout combination lamps, rear	Military type, MS51330	2 ea.
Reflectors	MS35387	8 ea., min.
Low tension cable	MIL-C-13486, 14 A.W.G. Min.	AR
Harness insulation	MIL-I-631 & MIL-I-15126	AR
Loom	Semitrailer type polyethylene tubing or oil proof SAE J562 loom	AR
Conduit	Commercial, thinwall type	AR
Cushioned support clamps	MS21322, MS21333 & MS21334	AR
Circuit number bands	MIL-B-43436 & MIL-B-43436/1	AR
Friction connectors	MS27142 through MS27148 & MS33800, waterproof	AR
Connector lubricant	MIL-S-8660	AR
Connector holder clip	Plastic or metal spring clip, similar to MIL-C-24066 & MIL-C-24066/2-329	AR
Terminal connectors	Spade type per SAE J561 with heat shrinkable or integral insulator sleeves	AR
Hazard warning signal	DOT Regulations, SAE J910, and SAE J945	1 assy

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3.23.1 Wiring. The wiring for both the service and blackout light systems shall be in compliance with the figure furnished by the contracting officer (see 6.3). The wiring shall be protected from road splash. Cushioned clamps, loom, thinwall conduit and grommets shall be provided to prevent whipping, chafing and other mechanical and heat damage. Wiring shall be fastened to the frame at intervals not to exceed 3 feet (91 cm). Two or more cables following the same routing shall be enclosed in loom or conduit having each end waterproofed, or shall be bound together in harness with 0.5 overlapping wraps of plastic electrical tape. Splices shall be waterproof. Mechanical splicing devices, if used, shall be fastened to the frame members with the interior of the device waterproofed with silicon sealant. All circuits shall be identified by number coding as specified in the government furnished diagrams (see 6.2). The circuit number bands shall be installed near each connection. The mating surfaces of the elastomer friction connectors shall be sealed with silicon sealant prior to assembly. The assembled friction connections shall be secured to the frame with holder clips to prevent cable fatigue breakage. Terminal connectors shall be of the ring type, and application shall be limited to use as component ground leads. The connections of this additional wiring to the wiring of the truck chassis-cab shall be made at the tail lamp friction connectors under the front fenders.

3.23.2 Lighting. All lamps shall operate as specified herein on 24 volts. The two lighting systems, service and blackout, shall be separate systems except for having common ground returns. The two front service turn signal lamps shall be symmetrically pedestal mounted on the top of the horizontal portion of the front fenders. The lamps shall be double faced, amber front and red rear lenses, having either sealed units of bulb and lenses in a nonferrous housing. If furnished, bulbs shall be of the single contact base type, trade No. 1683. Under fender boots, filled with silicon sealant, shall be furnished to waterproof the lamp. Friction retainment connectors shall be furnished on the pigtail lead. Each front lamp shall be equipped with brush guards and installed as shown in Government furnished diagrams (see 6.2). The service, clearance, sidemarker and identification lamps (MS35423) shall be amber forward and red rearward. One clearance lamp shall be mounted at each corner of the body and one service sidemarker lamp shall be mounted at each corner, for a total of eight lamps. Three service identification lamps shall be mounted on the front and on the rear, for a total of six lamps. At the rear, two service combination lamps shall be symmetrically flush or recess mounted, as far apart as practicable, within industry standard 4.5-inch (11.43 cm) diameter holes, to permit replacement of assemblies. These combination lamps shall be either sealed units inserted in elastomer rings, or bulb and lens mounted in an elastomer housing. If furnished, bulbs shall be double contact indexing base type, trade No. 1662. Grounding pigtail leads shall be required. Friction retainment connectors shall be furnished on the pigtail leads for tail and for stop and turn. The wires and forward side of these lamps shall be shielded to prevent damage from road splash and mud. One blackout clearance lamp and one blackout sidemarker lamp shall be mounted immediately below or alongside the corresponding service clearance and sidemarker lamps, for a total

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of eight. The colors of the blackout lamp lenses shall correspond to those of the service lamps. The two blackout combination lamps shall be mounted inboard or under the rear service combination lamps. These blackout lamps shall be centered and recessed within 6-inch (15.24 cm) diameter holes such that their lens doors are within 1-inch (2.54 cm) tolerance of the service combination lamp lenses. The blackout combination lamps shall be protected with guards equal to 6-inch (15.24 cm) diameter, schedule 40, steel pipe over at least the upper half of the lamp diameter.

3.23.3 Hazard warning operating unit. The hazard warning operating unit shall have a load capacity to flash all four truck service turn signal lamps, plus a minimum of two additional turn signal lamps of a trailer equipped with rear lamps having the same current draw as those furnished on the truck, in accordance with DOT Regulations. The hazard warning unit shall include a heavy-duty flasher, a multiple contact switch and an indicator light. The 24-volt flasher shall be waterproof, of either the electromechanical type or of the solid state type. The switch shall function with the truck's turn-signal-system which includes stop and turn circuits using the same lamp filament.

3.24 Spare tire carrier. The spare tire carrier shall be mounted on the side of the distributor. The carrier shall be mounted in an accessible location that shall permit removal and replacement of the spare tire and wheel using only the tools supplied with the truck. The carrier shall be mounted and braced to prevent damage to the tank or other parts of the distributor.

3.25 Fenders. The fenders for the rear wheels on each side of the truck shall be constructed of 11 gage, sheet steel. The fender clearance shall be determined on a level surface when the distributor is truck-mounted and the tank is filled. The fender clearance shall be not less than 11 inches (29.94 cm) above the chassis frame at centerline of the rear wheels, and not less than 5 inches (12.7 cm) between the rear of the rear tire and the fender. The rear mudguard portion of the fender shall be rigidly mounted, shall extend the full width of the tires, and shall extend downward to at least the centerline elevation of the tires. The front mudguard portion of the fender shall be formed by connecting the fender to the vertical sheet steel furnished on the truck chassis and located just to the rear of the fuel tanks. The fenders shall cover the width of the tires. The top of the fenders may serve as part of the catwalks.

3.26 Catwalks, steps, guard rails, and handrails. Catwalks and steps shall be fabricated from nonskid metal decking.

3.26.1 Catwalks. Full-length catwalks shall be provided on each side of the water tank. Guards shall be provided to prevent the use of the truck fuel tanks to obtain access to the catwalks. Clearance between the catwalk and the truck fuel tank shall permit emptying the contents of a can conforming to MIL-C-1283, equipped with a spout conforming to MIL-S-1285, into the fuel tank.

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3.26.2 Steps. Steps shall be provided on both sides of the distributor for access to the operator's platform, and when the catwalks and operator's platform are at different elevations, steps shall be provided between platform and catwalks. Steps shall be not less than 10 inches (25.4 cm) wide and shall be flanged. A ladder or rungs shall be provided on the tank for access from the catwalk to the manhole on top of the tank.

3.26.3 Guard rails. Guard rails shall be installed at a height of not less than 42 inches (106.68 cm) around the perimeter of the operator's platform. Intermediate rails shall be placed at 21 inches (53.34 cm) above the platform (see 3.9). Guard rails need not be installed at perimeter locations where distributor equipment restricts access by the operator. Guard rails shall have openings at step locations to provide access to and from the platform and catwalks, and to and from the ground and platform. Detachable chains shall be provided at openings.

3.26.4 Handgrasps. Steel pipe handgrasps shall be mounted on both sides of the tank and shall extend the full length of the tank. The handgrasps shall be placed 42 inches (106.68 cm) above the standing surface. In addition, handgrasps shall be provided on both sides of the operator's platform to aid personnel in mounting and dismounting from the operator's platform. The handgrasps shall be positioned so that personnel's hands can be maintained on the handgrasps at least at 54 inches (137.16 cm) above the standing surface while mounting or dismounting from the operator's platform. Clearance shall be provided to prevent personnel's hands from being caught between the handgrasps and any other distributor component.

3.27 Signal system. A signal system shall be provided. The signal system shall be of the mechanical, two-way, gong type and shall be operable from the truck cab and the operator's platform. The pulleys used to guide the cable to the gong shall permit operation of the gong without the cable riding off the pulley.

3.28 Government-furnished property. The following property will be furnished by the Government (see 6.3):

Item No.	Description	Identification	Quantity
1	Paint pattern	Camouflage patterns	As required (see 6.2.1)
2	Chassis drawings	Chassis type	As required (see 6.2.1)
3	Wiring diagrams	Wiring figures	As required (see 6.2.1)
4	6 x 6, 5-ton	Cab and chassis	As required (see 3.17)

3.29 Noise Limits. The noise produced by the distributer shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3 and 5.4, when tested in accordance with 4.5.2.11. The provision of MIL-STD-1474, 4.3 and 4.4, shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the

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procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.3, except the sign will state "HEARING PROTECTION REQUIRED AT OPERATING STATION." The sign shall be readable at the distance indicated on the sign.

3.30 Workmanship.

3.30.1 Steel fabrication. Steel used in the fabrication of the distributor shall be free from kinks and sharp bends. Steel plate shall not be warped. The straightening of material shall be done by methods that will not cause injury to the metal. Corners shall be square and true. Flame cutting may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined to remove ash and cooling checks. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape.

3.30.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided and all bolts, nuts and screws shall be tight.

3.30.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full and concentric with the rivet holes, and in full contact with the surface of the member.

3.31 Welders and welding.

3.31.1 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualifications is effective as defined by the particular code:

AWS D1.1 - Structural Welding Code - Steel.
ASME Boiler and Pressure Vessel Code.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Certification shall be made available for review, upon request, by the contracting officer or designated representative. The Government reserves the right to require the recertification of any welder or welding operator.

3.31.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing and other foreign matter. Parent materials, weld filler metals and fabrication techniques shall be as required to enable the distributor to conform to

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the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.187 (0.48 cm) inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the pump unit shall be the option of the contractor.

3.31.3 Surfaces. All parts, components and assemblies including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces and welded parts shall be cleaned and free of sand, dirt, fins, pits, sprues, scale, flux and other harmful or extraneous materials. External surfaces shall be smooth, and all edges shall be rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own, or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

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4.3 First article inspection.

4.3.1 First article inspection. The first article distributor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article distributor shall be tested as specified in 4.5.2.1 through 4.5.2.11. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each distributor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each distributor shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The distributor shall be examined as specified herein for the following defects:

101. Dimensions not as specified.
102. Parts or components missing, damaged, or do not function.
103. Assembly incomplete or not correct.
104. Material not as specified (see 3.3).
105. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
106. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
107. Contractor does not have documentation available for identification of material, material finishes, or treatment (see 3.3.1.2).
108. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the distributor (see 3.3.2).
109. Water tank not as specified (see 3.4).
110. Tank gage not as specified (see 3.4.1).
111. Pump unit not as specified (see 3.5).
112. Pump fittings and connections not as specified (see 3.5.1).
113. Pressure gage not as specified (see 3.6).
114. Spraybar not as specified (see 3.7).
115. Spraybar clearance not as specified (see 3.7.1).
116. Controls not located or identified as specified (see 3.8).
117. Operating instruction plate not as specified (see 3.8).
118. Operator's platform not as specified (see 3.9).
119. Speed-and-distance-indicating device not as specified (see 3.10).
120. Lubricants not as specified (see 3.11.1).

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121. Lubrication fittings not as specified (see 3.11.2).
122. Lubrication devices not accessible, missing, or not as specified (see 3.11.3).
123. Grease lubrication not as specified (see 3.11.4).
124. Oil lubrication not as specified (see 3.11.5).
125. Tiedown and lifting devices not as specified (see 3.12.1 and 3.12.2).
126. Accessories not as specified (see 3.13).
127. Suction hose assembly not as specified (see 3.13.1).
128. Discharge hose assembly not as specified (see 3.13.2).
129. Threads not as specified (see 3.14).
130. Repair parts or tools missing (see 3.15).
131. Toolbox not as specified (see 3.16).
132. Distributor mounting not as specified (see 3.17).
133. Treatment and painting not as specified (see 3.18).
134. Color not as specified (see 3.18).
135. Camouflage patterns not as specified (see 3.18.1).
136. Electromagnetic interference suppression components and network not identical with approved model (see 3.19).
137. Distribution chart not as specified (see 3.20).
138. Identification markings missing or not as specified (see 3.21).
139. Transportation data identification plates missing or not as specified (see 3.21.1).
140. Instruction and caution plates missing or not as specified (see 3.21.2).
141. Stenciling not as specified (see 3.22).
142. Electrical system not as specified (see 3.23).
143. Wiring system not as specified (see 3.23.1).
144. Lighting system not as specified (see 3.23.2).
145. Hazard warning operating unit not as specified (see 3.23.3).
146. Spare tire carrier not as specified (see 3.24).
147. Fenders not as specified (see 3.25).
148. Catwalks, steps, guard rails and handgrasps not as specified (see 3.26, 3.26.1, 3.26.2, 3.26.3 and 3.26.4).
149. Signal system not as specified (see 3.27).
150. Steel fabrication not as specified (see 3.30.1).
151. Bolted connections missing or not as specified (see 3.30.2).
152. Riveted connections missing or not as specified (see 3.30.3).
153. Contractor does not have documentation available for certification of welders (see 3.31.1).
154. Welding not as specified (see 3.31.2).
155. Surfaces not as specified (see 3.31.3).

4.5.2 Tests.

4.5.2.1 Interior coating. Interior coating thickness shall be tested using any suitable electronic or magnetic film thickness tester. A minimum of 10 random readings shall be made and shall include at least two readings on each

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interior surface. Any reading less than that specified in 3.4 shall be cause for rejection.

4.5.2.2 Tank capacity and tank gage accuracy. Fill the water tank approximately half full with water, read the tank gage and record the indicated volume of water. Determine and record the weight of the water contained in the tank. Then, completely fill the tank with water, read the tank gage and record the indicated volume of water. The actual volume of water for both the approximately-half-full and full conditions shall be determined by converting total weight of water to volume by use of unit weight. Indicated volume by tank gage not being within 25 gallons (94.63 liters) of actual volume of water or actual volume of water in a full tank of less than 1,000 gallons (3,785.3 liters) shall constitute failure of this test.

4.5.2.3 Application and distribution. Operate the truck-mounted distributor to apply water through the maximum spraybar width at rates of 0.8, 1.5, and 2.0 gallons per square yard (2.53, 4.75 and 6.33 liters per square meter) and through a 4-foot (1.22 m) spraybar at a rate of 0.1 gallon per square yard (0.32 liters per square meter) over a predetermined distance of 100 feet (30.48 m) for each of their rates and widths as follows:

a. Weigh truck-mounted distributor filled with water.

b. Apply the water over the 100-foot (30.48 m) distance at a required rate of application (speed of truck and pressure setting should be obtained from distribution chart specified herein).

c. Using a stopwatch, determine time required for truck-mounted distributor to apply water over the predetermined distance.

d. Weigh truck-mounted distributor.

e. Determine amount of water applied by difference in weight before and after test and convert to gallons.

f. Compute rate of application in gallons per square yard by utilizing amount of water applied, travel distance, application time, and spraybar width.

Observe spray pattern during all rates of application and observe applied water after each application for streaking or puddling. Observe spraybar for leaks throughout this test. With the spraybar at maximum width, operate the truck-mounted distributor at the speed required for an application rate of 0.1 gallon per square yard (0.475 l/m^2) and bring one end of the spraybar into contact with a rigid obstacle. Observe the horizontal swing breakaway feature. Then fold the spraybar, lock in the folded position, and apply water through the 8-foot (2.44 m) center section. Any of the following shall constitute failure of this test: Spray from any nozzle not extending to the centerline of adjoining

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nozzles; streaking or puddling of applied water; extended portions of the spraybar not swinging horizontally without damage to the spraybar or other parts of the distributor when extended portion strikes a rigid obstacle; the extended portions of the spraybar not being folded or locked in folded position as required; flow of water not being automatically cut off to the folded portions; rate of water actually applied not within plus or minus 5 percent of intended rate; extension section connections, spraybar end caps or parts of spraybar leaking; or forward speed for any distribution rate less than 88 fpm (26.82 mpm) nor greater than 1,350 fpm (411.48 mpm).

4.5.2.4 Operations. Perform the following operations:

a. Pump water from an external source into the distributor tank, utilizing all three 10-foot (3.048 m) sections of suction hose.

b. Pump water from distributor tank to each half of the spraybar separately.

c. Pump water from outside source to 1.5 inch (3.81 cm) outlet and through discharge hose assembly at maximum rate without water entering spraybar or tank.

d. Drain distributor tank by gravity through the spraybar.

Inability of the distributor to perform any of these operations in accordance with requirements specified herein shall constitute failure of this test.

4.5.2.5 Travel and mounting devices. Drive the fully loaded, truck-mounted distributor for 50 miles (30.47 km) over paved and gravel roads, with at least 30 miles (48.28 km) over gravel roads. The fully loaded, truck-mounted distributor shall then be subjected to a series of forward and reverse sudden stops as follows:

a. Accelerate forward from rest to 10 mph (16.09 Kmph) as quickly as possible, then apply brakes to stop suddenly. When forward motion has stopped, immediately accelerate to 10 mph (16.09 Kmph). The acceleration forward and sudden stop as described shall be accomplished 10 consecutive times.

b. Accelerate backward from rest to maximum reverse high-range speed, not to exceed 10 mph (16.09 Kmph), as quickly as possible, then apply the brakes to stop suddenly. When reverse motion has stopped, immediately accelerate to maximum high-range reverse speed, not to exceed 10 mph (16.09 Kmph). The reverse acceleration and sudden stop as described shall be accomplished 10 consecutive times.

Evidence of mounting devices permanently deformed or broken, alignment between distributor and chassis not being maintained, or accessories not remaining in their carrying places shall constitute failure of this test.

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4.5.2.6 Lifting and tiedown attachments. With the distributor anchored by means other than the provisions being tested, apply the required loads to each lifting and tiedown attachment as specified in MIL-STD-209. Record the dimensions of the attachment opening, the maximum load applied, and any nonconformance with this specification and MIL-STD-209. Any failure or permanent deformation to an attachment or its structural members shall constitute failure of this test.

4.5.2.6.1 Lifting attachments on 5-ton military truck chassis. Lift the complete truck-mounted distributor and hold in suspension in its normal travel position using slings that converge not more than 24 feet (7.3 m) above the lowest extremity of the distributor. Damage or failure to any part of the distributor, or sling clearance not maintained as specified, shall constitute failure of this test.

4.5.2.7 Rail impact test. With the distributor mounted on a 5-ton truck, and empty of water, subject the rail transport configuration to the rail impact test procedure described in MIL-STD-810. Permanent deformation or failure of any component of the distributor or truck (caused by the distributor) shall constitute failure of this test.

4.5.2.8 Speed-and-distance-indicating device. Drive the truck-mounted distributor over a measured distance of not less than 5,000 feet (1524 m) at an indicated speed of 500 fpm (152.4 mpm). Determine and record the time required to travel the measured distance at the indicated speed of 500 fpm (152.4 mpm) by a stop-watch. Record the trip-indicator reading at the start and end of the measured distance traveled. Indicated speed not being within 5 percent of actual speed, or indicated distance traveled not being within 5 percent of actual distance traveled shall constitute failure of this test.

4.5.2.9 Electromagnetic interference. The distributor shall be tested to determine conformance with 3.19. The supplier shall furnish a report showing permissible interference limits and the recorded level of interference. Nonconformance to the requirements of 3.19 shall constitute failure of this test.

4.5.2.10 Performance. Fill each distributor with water and operate for a run-in period of one hour. Check for leaks, demonstrate operation of all controls and make any adjustments. Any leak or malfunction of any component that cannot be corrected by adjustment shall constitute failure of this test.

4.5.2.11 Noise Level Test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position. Additionally, the

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noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packaging, packing and marking shall be examined to determine compliance with the applicable inspection requirements of MIL-D-771.

5. PACKAGING

5.1 Preservation and packing. Each complete distributor with components, accessories, repair parts, tools and technical publications shall be preserved and packed in accordance with MIL-D-771. Preservation and packing shall be level A, B, or C as specified (see 6.2).

5.2 Marking. Boxed components, shipping containers and mobile distributors shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The water distributors are primarily intended for use in soil stabilization and soil compaction operations requiring the accurate distribution of water at predetermined rates. The water distributors can also be used for transporting or transferring water, for application of water for dust alleviation, for sump pumping, for washing equipment and as auxiliary fire fighting equipment.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Time frame required for submission of first article model (see 3.2).

c. Color required when other than specified (see 3.18).

d. When specified, time frame when camouflage pattern data is required (see 3.18.1).

e. Repair parts, maintenance tools and operating tools required (see 3.15).

f. When distributor is to be furnished unmounted (see 3.17 and 6.5).

g. Levels of preservation and of packing required (see 5.1).

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h. Type of chassis required (see 3.17).

i. Wiring diagram required (see 3.23.1).

j. When specified, welding process used in fabrication of the pump unit (see 3.31.2).

6.2.2 Data requirements. When this specification is used in an acqusition which incorporate a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirements	Applicable DID No.
3.18	Color, Photograph Prints	DI-MISC-80192
3.18.1	Camouflage, Line Art Data	DI-MISC-80176

6.3 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.28. The contracting officer should specify the chassis that will be furnished.

6.4 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development and Engineering Center (ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.5 Unmounted distributors. When unmounted distributors are specified, quality conformance test (see 4.4) should be run on static-mounted distributor and the preproduction model should be furnished mounted on the Government-furnished chassis specified.

6.6 First article. When a first article inspection is required, the items should be a first article model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and

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approval of the first article test results and disposition of the first article.

6.7 Subject term (key word) listing.

Self-loading
Soil compaction
Soil stabilization
Spray, adjustable
Tank-type
Transporting
Truck mounted

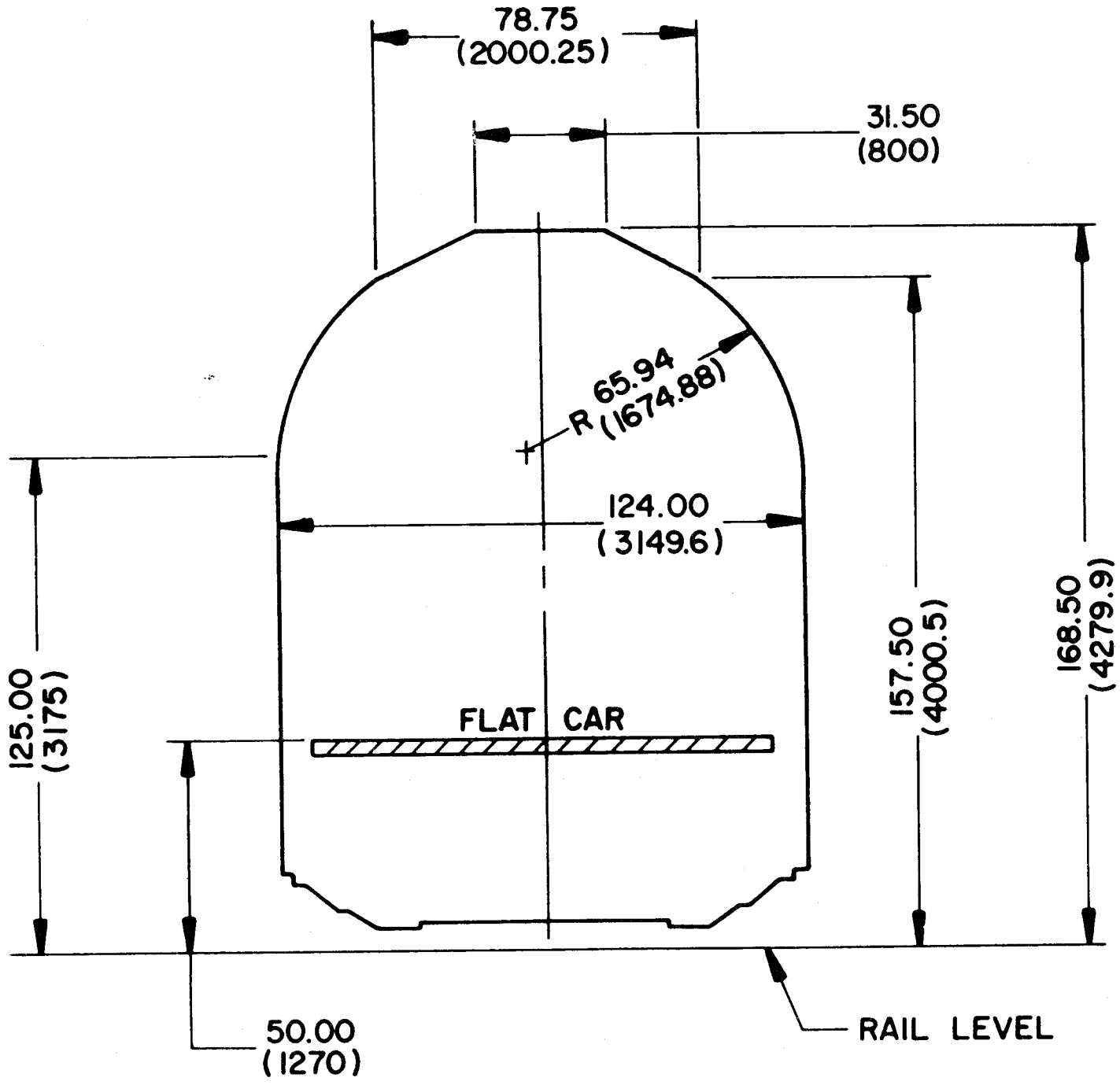
Custodian:
Army - ME

User interests:
Army - CE, AT

Preparing activity:
Army - ME

Project No. 3825-A179

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NOTE: DIMENSIONS SHOWN: INCHES (MILLIMETERS)

FIGURE 1. Passe-partout international (PPI) gauge.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-13687F(ME)	Distributors, Water, Tank-Type: Truck Mounted

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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